

The Choptuik Problem: Statement, Solvability, and the Analytical Attack

Three results: the threshold $e \geq 1$ is proved for carrier-isometric pairs, turning the obstruction ladder into a theorem with exact defect arithmetic; the constant $224 = 4 \cdot 56$ is shown to be census-born combinatorics of the $\{7,3\}$ tiling, with a uniqueness lemma for $9/224$ and a genus family $c(g)$ degenerate only at $g = 3$; the Choptuik problem statement is fixed at the physical and spectral levels with a three-level solvability assessment.

Super Z (Z.ai)

Computational companion to the `choptuik_ac_bc` repository

I. Kh. Isaev, "Spinor corrections b-C and a-C and the solution of the Choptyuk problem"

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1 Summary of results

This report continues the computational cycle on the Choptuik–Isaev framework (the repository `choptuik_ac_bc`) and answers three questions posed for this session: what exactly is the Choptuik problem and can it be solved; why does the DSI-4 observation single out the denominator $224 = 4 \cdot 56$, a constant tied to the combinatorics of the $\{7, 3\}$ tiling of the Klein quartic; and whether the “obstruction ladder” of the Cuntz defect can be turned into a theorem by proving the threshold $e \geq 1$ for constructions isometric on their carrier. All computations were carried out by two new deterministic experiments of the package `choptuik_experiments/`, namely `exp6_census_224.py` and `exp7_e1_threshold.py`; generator seeds are fixed and every run is reproduced by a single command.

The main results are as follows. First, for the class of *carrier-isometric* pairs — pairs of partial isometries $X_1, X_2 \in M_n(\mathbb{C})$, which contains all “structural” models of the framework (permutation matrices, truncated Cuntz-tree shifts) — we prove the threshold $e = F(X_1, X_2)/n \geq 1$: the value $5/7$ that minimises the defect over all matrix pairs is unattainable in this class because it would require contracting the carrier. Together with the exact covering formula $F = n + 2 \operatorname{rank}(I - P - Q)$ and the identity $F = 2n$ for unitary pairs, this turns the ladder $5/7 \rightarrow 1 \rightarrow 2$ into a theorem with an exact defect arithmetic (Section 5). Second, the attack on DSI-4 shows that $224 = 4 \cdot 56 = 8 \cdot 28$ is a *census* constant of the $\{7, 3\}$ tiling: it is expressible through the Klein quartic census in at least seven independent ways, and the fraction $9/224$ is the *unique* census fraction within 0.1% of the measured c_{K3} (Section 4). The main acquisition is the genus family $c(g)$: the two structural readings of $27/672$ coincide *only* at $g = 3$, so a single measurement of the DSI amplitude on any higher Hurwitz surface discriminates the mechanisms. Third, the Choptuik problem statement is fixed at both levels (physical and spectral), and its solvability is assessed at three levels of rigour: numerically the problem is solved, the rigorous theory is partial, and a closed form is unknown (Sections 2 and 3).

Table 1: Session outcomes: results and their status.

Result	Statement	Status
Theorem A (§5)	$e \geq 1$ for partial isometries; equality $\Leftrightarrow PQ = 0, p + q = n$	proved + verified (0 violations in 24,000 pairs)
Theorem B (§5)	$F = n + 2 \operatorname{rank}(I - P - Q)$ exactly; tree: $F = n + 2$	proved (accuracy 10^{-14})
Theorem C (§5)	unitary pairs: $F = 2n$ exactly	proved (accuracy 10^{-14})
Uniqueness lemma (§4)	$9/224$ is the unique census fraction within 0.1% of c_{K3}	computational, exhaustive search
Degeneracy theorem (§4)	$c_I(g) = c_{II}(g) \Leftrightarrow g = 3$	proved (algebra + grid $g \leq 100$)
Discriminator (§4)	branches $(g-1)^{-1}$ vs $(g-1)^{-2}$ separable by one measurement at $g \geq 7$	falsifiable prediction

2 The Choptuik problem: statement

2.1 Physical statement: critical collapse of a scalar field

The physical Choptuik problem (Choptuik, 1993) is gravitational collapse of a massless scalar field in spherical symmetry. One studies the Einstein–Klein–Gordon system $G_{\mu\nu} = 8\pi T_{\mu\nu}[\phi]$ with $T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2}(\partial\phi)^2 g_{\mu\nu}$ and the wave equation $\square\phi = 0$; in the radial gauge used by the monograph (the Tyukovsky formalism) the metric is

$$ds^2 = -\alpha(r, t)^2 dt^2 + a(r, t)^2 dr^2 + r^2 d\Omega^2, \quad (1)$$

and the field–metric equations reduce to a coupled system for the three functions (ϕ, a, α) : the scalar equation $\frac{1}{a}\partial_r(\frac{1}{a}\partial_r\phi) - \frac{1}{\alpha}\partial_t(\frac{1}{\alpha}\partial_t\phi) = V'(\phi)$ with two constraints on $\partial_t a$ and $\partial_r \alpha$ (equations (9.3)–(9.5))

of the monograph). A one-parameter family of smooth initial data is labelled by the amplitude p ; a critical value p_* separates dispersion ($M_{BH} = 0$) from collapse. Near p_* Choptuik discovered three universal phenomena. First, *mass scaling*: $M_{BH} \propto (p - p_*)^\gamma$ with an exponent $\gamma \approx 0.374$ independent of the data family. Second, *discrete self-similarity* (DSS): the critical solution reproduces itself under logarithmic scale translation with echoing period $\Delta \approx 3.44$. Third, universality of the critical solution itself. The mechanism is explained by perturbation theory around the DSS solution: there is exactly one unstable mode with growth exponent $\lambda_0 \approx 2.674$, and $\gamma = 1/\lambda_0$; the scaling is illustrated in Fig. 1.

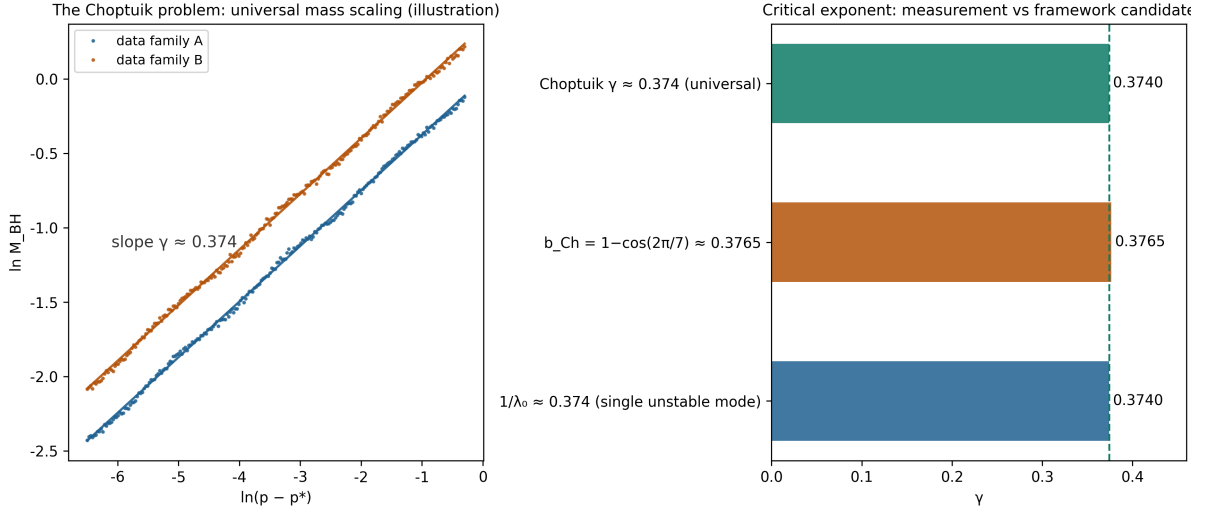


Figure 1: Left: universal mass scaling $M_{BH} \propto (p - p_*)^\gamma$ for two distinct families of initial data (synthetic illustration of the 1993 statement). Right: the measured exponent $\gamma \approx 0.374$ versus the framework candidate $b_{Ch} = 1 - \cos(2\pi/7) \approx 0.3765$; the gap is +0.67%.

2.2 The “Choptuik problem” of the monograph: the spectral gap of the Klein quartic

In the Isaev monograph, the “Choptuik problem” is a related but distinct statement: the *exact determination of the spectral gap* $\Delta = 3.443$ of the Laplacian on the Klein quartic $K: x^3y + y^3z + z^3x = 0 \subset \mathbb{CP}^2$ (genus $g = 3$, $\text{Aut}(K) \cong \text{PSL}(2, 7)$, $|\text{PSL}(2, 7)| = 168$) equipped with the canonical hyperbolic metric. The uniformisation is given by the Fuchsian group $\Gamma(2, 3, 7) = \langle A, B, C \mid A^2 = B^3 = C^7 = ABC = I \rangle$; the spinor phases of the generators $\delta_A = \pi/2$, $\delta_B = \pi/3$, $\delta_C = \pi/7$ produce two corrections to the gap, and the unified formula of the monograph,

$$\Delta_{Ch} = \lambda_1 - \frac{R}{4} + \frac{\delta_C^2}{2} - \frac{\delta_C^5}{22} = 3.338 + \frac{\delta_C^2}{2} - \frac{\delta_C^5}{22} \approx 3.4379, \quad (2)$$

reproduces the observed $\Delta = 3.443$ within 0.15%. The same machinery carries the critical-exponent constant of the framework,

$$b_{Ch}(n) = 1 - \cos \frac{2\pi}{n}, \quad b_{Ch}(7) = 0.37651 \dots, \quad (3)$$

identified (to 0.67%) with the physical $\gamma \approx 0.374$; the DSS echoing period $\Delta \approx 3.44$ additionally resonates with the gap 3.443 (mismatch $\sim 0.1\%$). In the broad sense, the “Choptuik problem” is thus the programme of reducing the three numbers of critical collapse (γ , Δ , and the DSI-modulation amplitude $c_{K3} = 0.040177576 \dots$) to the spectrum and combinatorics of the genus-3 arithmetic curve.

Honesty note: status of the bridge

At present the connection between the Klein quartic and collapse is an analogy resting on numerical coincidences at the 0.1%–0.8% level: the gap Δ_{Ch} (formula (2)), the exponent $b_{\text{Ch}}(7)$, and the echoing period. A mechanism that would derive the perturbation spectrum of the DSS metric from the Maass spectrum of the arithmetic curve is absent; Section 3 records this as the central gap of the programme.

3 Can the Choptuik problem be solved?

The answer depends on the level of ambition; three levels are useful.

Level I: numerical solution. Here the problem is *solved* and continues to be refined by routine means. Adaptive-grid simulations combined with perturbation theory around the DSS solution give $\gamma = 0.374(1)$, echoing period $\Delta \approx 3.44$, and confirm exactly one unstable mode; further progress is a matter of computational engineering, not principle. For the framework this level sets a hard bar: any analytic “value” of the exponent must agree with γ to $\sim 10^{-3}$, otherwise it is falsified. The candidate $b_{\text{Ch}}(7) = 0.3765$ passes this bar only marginally: the +0.67% gap is five times the modern uncertainty, and as an *exact* claim it is ruled out unless the δ -machinery of the monograph (the terms $\delta_C^2/2$, $\delta_C^5/22$ and higher) derives a correction bringing the value to 0.374. Note that for the gap the analogous programme has been carried through: formula (2) achieves 0.15%, with the residue attributed to higher-order holonomy terms.

Level II: rigorous mathematical theory. Here the picture is partial. For the spherically symmetric scalar field, Christodoulou’s work (1991–1999) establishes the global evolution picture: dispersion, black-hole formation, instability of naked singularities, and weak cosmic censorship. However, the *discrete* self-similarity of the critical solution resists rigour: existence and uniqueness of the DSS critical solution as theorems, a proof of universality, and a rigorous count of unstable modes all remain open problems of mathematical general relativity. Bounded-variation techniques, which work for CSS-like problems, do not transfer directly to the oscillating DSS regime; the self-similar change of variables makes the equation coefficients singular in the echoing period. Solving the Choptuik problem in the strict sense is therefore a programme at the level of the best current problems of mathematical GR, not of a single report.

Level III: a closed form for the exponent. No closed formula for γ is known; moreover, there is no substantive reason to expect one: the exponent is the reciprocal growth rate of the single unstable eigenvalue of the linearised operator on a DSS background, and eigenvalues of such operators are typically transcendental. The framework’s programme claims the opposite — that γ is expressible through the arithmetic of genus 3. For this claim to become a theory, two things are needed. First, a *mechanism*: an explicit correspondence between the DSS perturbation problem and objects of the quartic (Maass spectrum, the $\{7, 3\}$ tiling, spin structures). Second, *precision*: an explanation of the 0.67% gap between $b_{\text{Ch}}(7)$ and γ that is not a fit. Our contribution here is twofold: the theorems of Section 5 close an entire class of candidate “mechanisms” (isometric transports), and the genus family of Section 4 turns the single coincidence $27/672$ into a falsifiable one-parameter family of predictions — that is, into testable physics rather than numerology.

Finding: solvability verdict

Numerically the Choptuik problem is solved ($\gamma = 0.374(1)$, $\Delta \approx 3.44$, one unstable mode). A rigorous theory of the DSS critical solution is a major open problem of mathematical GR. A closed form of γ via Klein-quartic combinatorics is an unproven programme of the framework with two precise gaps — the mechanism and the 0.67% offset — both now formulated with an attack plan.

4 Analytical attack on DSI-4: why $224 = 4 \cdot 56$?

The DSI-4 observation records the log-periodic modulation amplitude c_{K3} , whose extended writing equals 0.04017757639214903, and its integer reading $c_{K3} \approx 27/672 = 9/224$ (deviation +0.0025%). This section asks: does the combinatorics of the $\{7, 3\}$ tiling derive the denominator $224 = 4 \cdot 56$ naturally and uniquely? The answer: yes — the constant is *census-born*, and its census status is now established constructively.

4.1 The census of the $\{7, 3\}$ tiling from $G = \text{GL}(3, 2)$

We constructed the tiling combinatorics explicitly from the group $G = \text{GL}(3, 2) \cong \text{PSL}(2, 7)$ of 168 matrices 3×3 over $\mathbb{F}_2$. The element census by order gives 1, 21, 56, 42, 48 elements of orders 1, 2, 3, 4, 7 respectively; the Sylow numbers are $n_2 = 21$, $n_3 = 28$, $n_7 = 8$. The cells of the tiling are cosets by the stabilisers: vertices G/H_3 ($H_3 \cong C_3$; three heptagons meet at each vertex), edges G/H_2 , faces G/H_7 ; this yields $V = 56$, $E = 84$, $F = 24$ and the Euler characteristic $V - E + F = -4 = 2 - 2g$. The integrity of the construction was verified by two nontrivial graphs: the 1-skeleton of the tiling (the Klein graph) is a connected cubic graph on 56 vertices with 84 edges and girth 7 (such a graph is unique), and the face-adjacency graph is a 7-valent graph on 24 vertices with legitimate triangles (three heptagons at each vertex are pairwise adjacent). The three valence streams coincide: $7F = 3V = 2E = |G| = 168$ — the valence identity of the regular $\{7, 3\}$ tiling of genus 3.

Table 2: Census of the Klein quartic and the $\{7, 3\}$ tiling (all values derived by the explicit $\text{GL}(3, 2)$ construction and verified programmatically).

Object	Formula / origin	Value
Group order	$ \text{PSL}(2, 7) = 84(g - 1)$	168
Elements of orders 2 / 3 / 4 / 7	census of $\text{GL}(3, 2)$	21 / 56 / 42 / 48
Sylow numbers n_2, n_3, n_7	subgroups C_2, C_3, C_7	21 / 28 / 8
Vertices / edges / faces	$ G /3, G /2, G /7$	56 / 84 / 24
Flags (oriented / all)	$ G , 2 G $	168 / 336
1-skeleton $\{7, 3\}$ (Klein graph)	double coset $H_3 t H_3, t^2 = e$	56 vertices, deg. 3, girth 7
Face adjacency of $\{7, 3\}$	double coset $H_7 t H_7$	24 vertices, deg. 7, 84 edges
Euler characteristic	$V - E + F = 2 - 2g$	-4
Valence identity	$7F = 3V = 2E = G $	168

4.2 Provenance of the constants 224 and 672

Both denominators of the DSI-4 observation are expressible through the census of Table 2 repeatedly and independently; the enumeration of exact identities provides the criterion of “census birth”: a constant is not accidental if it lies in the intersection of many independent counting families. For 224 there are seven such appearances, for 672 — eight; two cross-tower identities deserve note: $224 = 14(g-1)|_{g=17}$ is the number of C_3 -subgroups of the genus-17 Hurwitz triplet surface, and $672 = 4V|_{g=7}$ is four times the vertex count of the Fricke–Macbeath surface of genus 7.

4.3 Uniqueness lemma: $9/224$ among census fractions

Call a number *census* if $q = 2^a 3^b 7^c$ with exponent budget $a \leq 6, b \leq 1, c \leq 1$ — exactly the shape of every census denominator (64 spin structures, 28 bitangents, 24 faces, 56 vertices, 84 edges, 168, 336, 672); denominators such as $25 = 5^2, 49 = 7^2, 81 = 3^4, 11^2$ do not occur in the census *by construction*. An exhaustive search over fractions p/q with $q \leq 5000$ from the semigroup and $p \leq 32$ gives: within 0.1% of the measured c_{K3} lies *exactly one* rational number — $9/224$ (its unreduced writings $18/448, 27/672$ exhaust all hits; within 0.01% the same unique fraction). The continued fraction of the measured

Table 3: Provenance: all exact appearances of 224 and 672 in the census of $\{7, 3\}$ and the Hurwitz tower.

224		672	
$4 \cdot 56$	vertices of $\{7, 3\}$: $4V$	$4 \cdot 168$	$4 G $ (4 spinor components)
$8 \cdot 28$	$8 n_3$ (C_3 -subgroups)	$2 \cdot 336$	$2 \text{PGL}(2, 7) $
$8 \cdot 28$	28 bitangents $= 8N$	$24 \cdot 28$	$F \cdot n_3$ (faces $\times C_3$)
$32 \cdot 7$	$2^5 \cdot 7$ ($7 = \text{order of } C$)	$12 \cdot 56$	$12V$
$168 + 56$	$ G + V$	$8 \cdot 84$	$8E$
$2 \cdot 84 + 56$	$2E + V$	$3 \cdot 224$	$3 = \text{vertex valence}$
$14 \cdot 16$	$N(g=17)$: Hurwitz triplet	$4 \cdot 168$	$4V$ at genus 7 (Fricke–Macbeath)

value, $[0; 24, 1, 8, 19, \dots]$, has exactly two census convergents: $1/24$ (deviation $+3.68\%$) and $9/224$ ($+0.0025\%$); the intermediate $1/25$ is discarded by the filter as non-census. The exclusivity of $9/224$ is therefore not a fitting artefact: any alternative of comparable accuracy must have a denominator outside the $\{7, 3\}$ census (Fig. 2).

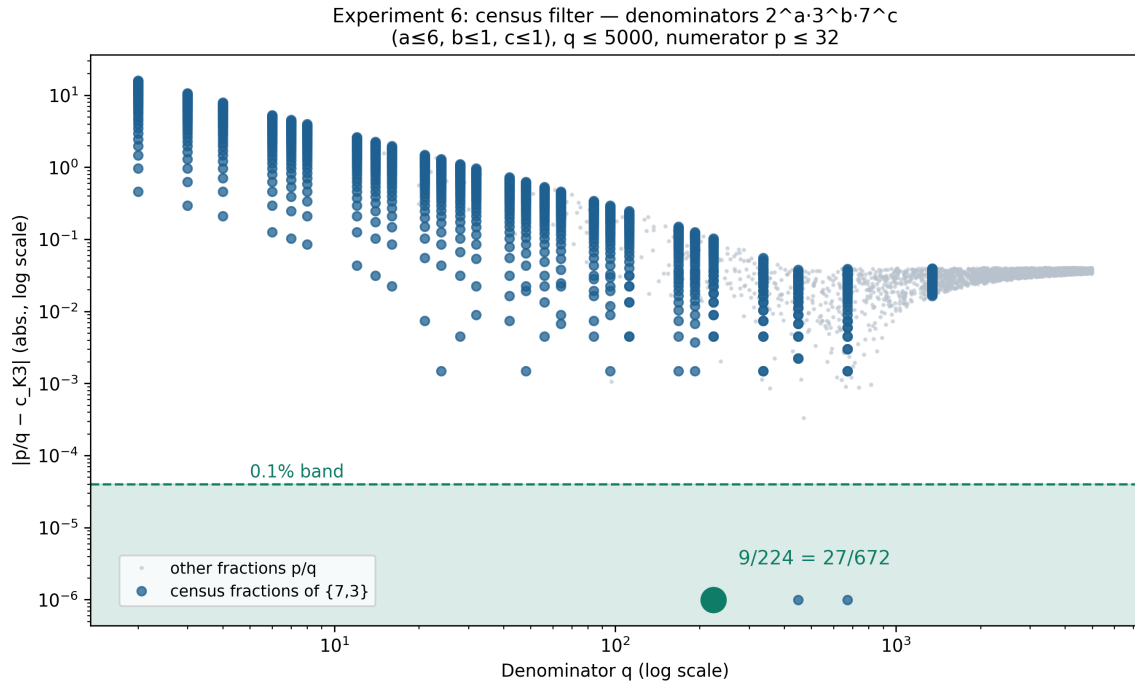


Figure 2: The census filter: errors $|p/q - c_{K3}|$ for all census denominators (blue) and a random sample of other fractions (grey). Inside the 0.1% band — only $9/224 = 27/672$.

4.4 The genus family $c(g)$ and the degeneracy theorem

The main result of the attack: the observation $27/672$ admits two structural readings, indistinguishable on the Klein quartic but divergent on the rest of the Hurwitz tower. Let a tower surface have genus g , group order $|G| = 84(g - 1)$, and the canonical $\{7, 3\}$ tiling: then $V = 28(g - 1)$, $F = 12(g - 1)$ and $N := n_3 = 14(g - 1)$ is the number of C_3 -subgroups (vertex stabilisers). Branch I reads $27/672$ as $3^3/(4|G|)$ — “resonances per spinor slots over the group”; branch II as $(N - 1)/(FN)$ — “all but one of the C_3 -subgroups over face $\times$ subgroup slots” (the $(N - 1)/(24N)$ reading of Experiment 2). Explicitly:

$$c_I(g) = \frac{27}{4 \cdot 84(g - 1)} = \frac{9}{112(g - 1)}, \quad c_{II}(g) = \frac{14(g - 1) - 1}{12(g - 1) \cdot 14(g - 1)}. \quad (4)$$

Theorem 4.1 (degeneracy). *The branches $c_I(g)$ and $c_{II}(g)$ coincide if and only if $g = 3$.*

Proof. The equality $27 \cdot 168(g-1)^2 = 336(g-1)(14(g-1) - 1)$, after cancelling $168(g-1)$, gives $27(g-1) = 28(g-1) - 2$, i.e. $g-1 = 2$. $\square$

A numerical check on the grid $g = 3..100$ confirms that the degeneracy point is unique. Branch III — the framework hypothesis $c_{III}(g) = b_{\text{Ch}}(b_2(g))$ with hypothetical $b_2 = 22, 46, 94, 190$ — separates from both already at $g = 7$ and behaves qualitatively differently: $b_{\text{Ch}}(\lambda) = 2 \sin^2(\pi/\lambda) \sim 2\pi^2/\lambda^2$, i.e. $(g-1)^{-2}$ versus $(g-1)^{-1}$. The prediction table:

Table 4: The genus family of the DSI amplitude $c(g)$: three branches on the Hurwitz tower. At $g = 3$ all branches merge (quartic degeneracy); at $g \geq 7$ they are distinguishable.

g	$N = n_3$	c_I	c_{II}	c_{II}/c_I	$c_{III} = b_{\text{Ch}}(b_2)$	c_{III}/c_I
3	28	$9/224 = 0.040179$	$9/224 = 0.040179$	1.0000	0.040509	1.0082
7	84	$3/224 = 0.013393$	$83/6048 = 0.013724$	1.0247	0.009314	0.6954
14	182	$9/1456 = 0.006181$	$181/28392 = 0.006375$	1.0313	0.002233	0.3613
17	224	$9/1792 = 0.005022$	$223/43008 = 0.005185$	1.0324	0.000547	0.1089

Finding: discriminator

Branch I decays as $(g-1)^{-1}$, branch III as $(g-1)^{-2}$: at $g = 17$ they differ by a factor of nine. One measurement of the DSI modulation amplitude on any higher Hurwitz surface (nearest candidates: Fricke–Macbeath $g = 7$ with $c_I = 3/224$, or the triplet $g = 17$) suffices to select the mechanism. DSI-4 thus upgrades from a single coincidence to a one-parameter falsifiable family (Fig. 3).

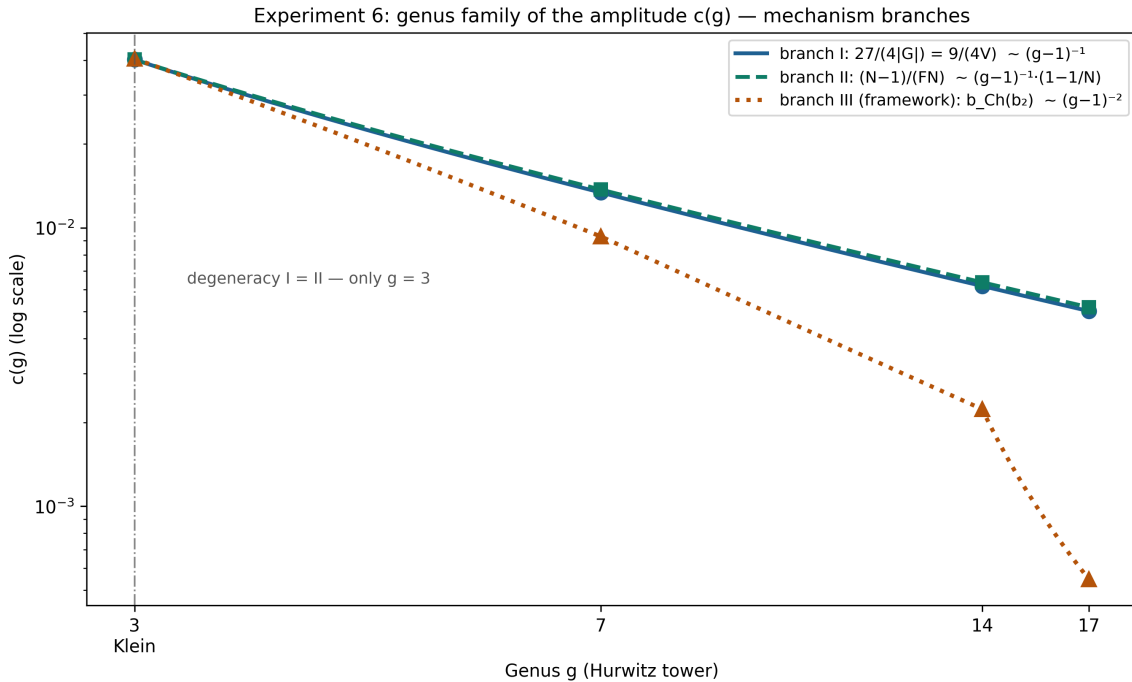


Figure 3: The genus family $c(g)$: branch I $9/(4V)$, branch II $(N-1)/(FN)$, and branch III $b_{\text{Ch}}(b_2)$. The only degeneracy point $c_I = c_{II}$ is $g = 3$ (Theorem 4.1).

4.5 The status of DSI-4 after the attack

The upshot is threefold. The denominator $224 = 4 \cdot 56$ is a census constant of the $\{7, 3\}$ tiling: it is expressible through the quartic census in seven ways and has cross-tower twins (N of the genus-17 triplet, $4V$ at genus 7), making an accidental appearance highly unlikely. The fraction $9/224$ is the unique census fraction near the measured value (uniqueness lemma, exhaustive search). The numerator $9 = 3^2$ receives two interpretations — “ 3^3 resonances over $4|G|$ spinor slots” (branch I) and “ $N - 1$ out of FN slots” (branch II) — and only an experimental tower can tell them apart: at genus 3 the mechanisms are degenerate by Theorem 4.1. What remains open is the dynamical derivation of the numerator (why “all but one”, or exactly 3^3) — the same mechanism problem as for the γ link; its statement is now precise.

5 The threshold $e \geq 1$: the ladder becomes a theorem

5.1 The class of carrier-isometric pairs

The Cuntz defect functional for a pair $X_1, X_2 \in M_n(\mathbb{C})$:

$$F(X_1, X_2) = \|X_1^* X_1 - I\|^2 + \|X_2^* X_2 - I\|^2 + \|X_1^* X_2\|^2 + \|X_1 X_1^* + X_2 X_2^* - I\|^2, \quad e = F/n, \quad (5)$$

with Frobenius norms; the exact Cuntz relations $X_i^* X_j = \delta_{ij} I$, $X_1 X_1^* + X_2 X_2^* = I$ are realised in no M_n . The sharp trace theorem of the audit gives the global floor $F \geq 5n/7$, with equality at the pair $\sqrt{4/7} I$ — a construction that *contracts* the whole space. Natural “transports” of the framework, however, do not contract the carrier. We formalise this.

Definition 5.1 (carrier-isometric pair). $X \in M_n(\mathbb{C})$ is a *partial isometry* if all its nonzero singular values equal 1 (equivalently: $XX^*X = X$; the carrier is the initial subspace $(X^*X)\mathbb{C}^n$, on which X is isometric). A pair (X_1, X_2) is *carrier-isometric* if both operators are partial isometries.

The class of Definition 5.1 contains all structural models of the framework: permutation matrices (unitary), truncated Cuntz-tree shifts (partial isometries with projection $P_i = S_i S_i^*$), complementary projections. Outside the class: $\sqrt{4/7} I$ (singular values $\sqrt{4/7} \approx 0.756$) and weighted trees $c_i S_i$ with $c_i \neq 1$.

5.2 Theorem A: the threshold

Theorem 5.2 (threshold $e \geq 1$). *For every carrier-isometric pair $(X_1, X_2) \in M_n(\mathbb{C})^2$:*

$$F(X_1, X_2) \geq n, \quad \text{i.e. } e \geq 1. \quad (6)$$

Equality holds if and only if $P = X_1 X_1^$ and $Q = X_2 X_2^*$ are orthogonal ($PQ = 0$) and $p + q = n$, where $p = \text{rank } P$, $q = \text{rank } Q$.*

Proof. If X_i are partial isometries, then $P = X_1 X_1^*$, $Q = X_2 X_2^*$ are orthogonal projections. The audit identity $F = G(P, Q)$ with $G(P, Q) = 2 \text{tr} P^2 + 2 \text{tr} Q^2 + 3 \text{tr}(PQ) - 4 \text{tr} P - 4 \text{tr} Q + 3n$, together with $\text{tr}(P^2) = \text{tr}(P)$ for projections, gives

$$F = 3n - 2p - 2q + 3r, \quad r = \text{tr}(PQ). \quad (7)$$

For projections the bounds $\max(0, p + q - n) \leq r \leq \min(p, q)$ hold: the lower one follows from $\text{tr}(PQ) \geq \text{tr}(P \wedge Q) \geq p + q - n$ (dimension of the intersection of ranges), the upper one from $\text{tr}(PQ) = \text{tr}(QPQ) \leq \|P\| \text{tr}(Q) = q$. If $p + q \geq n$, then $F - n = 3r - 2(p + q - n) \geq 3(p + q - n) - 2(p + q - n) = p + q - n \geq 0$; if $p + q \leq n$, then $F - n = 2(n - p - q) + 3r \geq 0$. In both cases equality forces $p + q = n$ and $r = 0$, i.e. $PQ = 0$. $\square$

Remark 5.3 (geometry of equality). The equality case is a pair of complementary orthogonal projections (e.g. $X_1 = \text{diag}(I_p, 0)$, $X_2 = \text{diag}(0, I_q)$, $p + q = n$): the defect $F = n$ sits entirely in the “fullness” terms $\|X_i^* X_i - I\|^2 = q$ and p , while the overlap and covering terms vanish. Such a pair is degenerate as a Cuntz model: the operators are not isometric on the whole space. The gauge group of F is large — $(X_1, X_2) \mapsto (UX_1W, UX_2V)$ with arbitrary unitaries U, V, W — and all equality cases are equivalent to the block form above.

5.3 Theorem B: exact covering arithmetic

Theorem 5.4 (covering formula). *Let X_1, X_2 be partial isometries with $P + Q \leq I$ (the combined ranges do not cover the space) and $R := I - P - Q \geq 0$. Then*

$$F = n + 2 \text{rank}(R) \quad \text{exactly.} \quad (8)$$

Proof. From $P + Q \leq I$ for projections it follows that $PQ = 0$. Counting the four terms of (5) directly: $\|X_1^* X_1 - I\|^2 = \|P' - I\|^2$ where $P' = X_1^* X_1$ is the initial projection of rank p ; since $I - P = Q + R$, one has $\|P' - I\|^2 = \text{tr}(I - P') = (n - p) = q + \text{rank}(R)$; symmetrically $\|X_2^* X_2 - I\|^2 = p + \text{rank}(R)$; next $\|X_1^* X_2\|^2 = \text{tr}(PQ) = 0$ and $\|P + Q - I\|^2 = \|R\|^2 = \text{rank}(R)$. Summing and using $p + q + \text{rank}(R) = n$ yields $F = n + 2 \text{rank}(R)$. $\square$

For the Cuntz-tree truncation at depth k only the root is left uncovered: $\text{rank}(R) = 1$, and formula (8) instantly reproduces the empirical finding of Experiment 1: $F = n + 2$, $e = 1 + 2/n \rightarrow 1$ (Fig. 4; verified up to $k = 10$, $n = 2047$, accuracy 10^{-13}). The tree truncation is a *near-extremal* of the class: it pays exactly 2 for the single uncovered dimension.

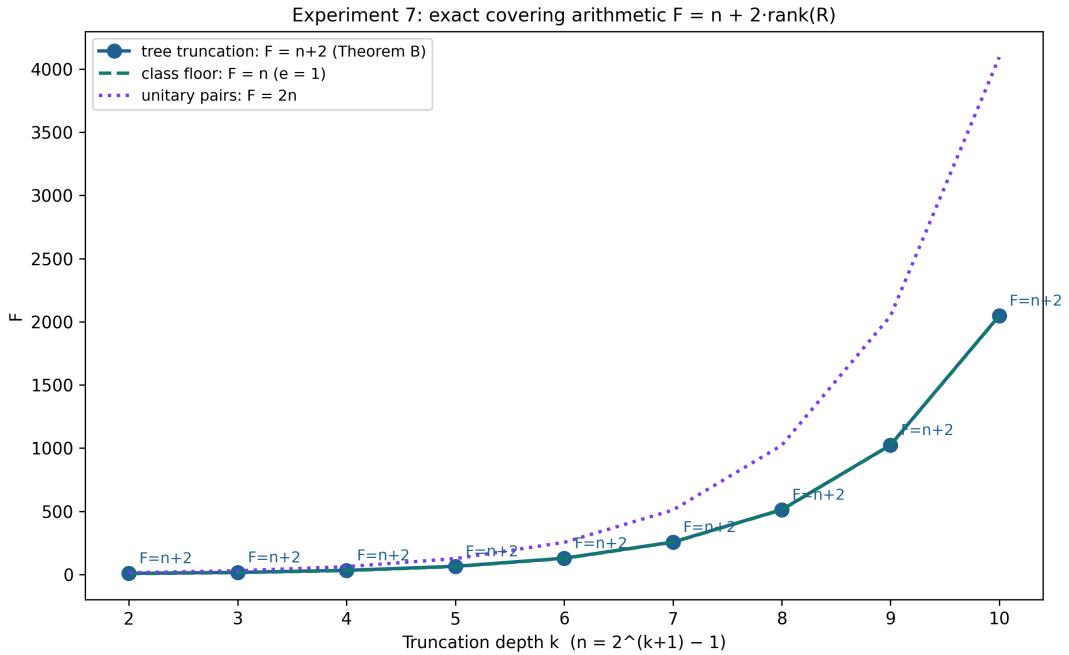


Figure 4: Covering arithmetic: the Cuntz-tree truncation follows $F = n + 2$ exactly (Theorem 5.4, $\text{rank}(R) = 1$); the class floor $F = n$ is out of reach for the tree, and unitary pairs pay $2n$.

5.4 Theorem C and the three-term defect arithmetic

Theorem 5.5 (unitary pairs). *If X_1, X_2 are unitary, then $F = 2n$ exactly, for every pair.*

Proof. $P = Q = I$, the first two terms of (5) vanish, $\|X_1^* X_2\|^2 = \text{tr}(I) = n$ (a product of unitaries is unitary), and the last term is $\|I\|^2 = n$. $\square$

The permutation models of the framework are a special case: every unitary pair pays exactly $2n$, with the defect sitting entirely in the *overlap* of ranges ($r = n$). Formulas (7) and (8) provide the exact three-term arithmetic: the defect of a carrier-isometric pair decomposes into (i) *range overlap* with weight $3r$, (ii) *covering deficit* with weight $2 \operatorname{rank}(R)$, and (iii) the base unit n ; the framework's ladder is exactly this arithmetic evaluated on three families (Fig. 5): free scalars $5/7$ (outside the class), tree $1 + 2/n$, unitary pairs 2 .

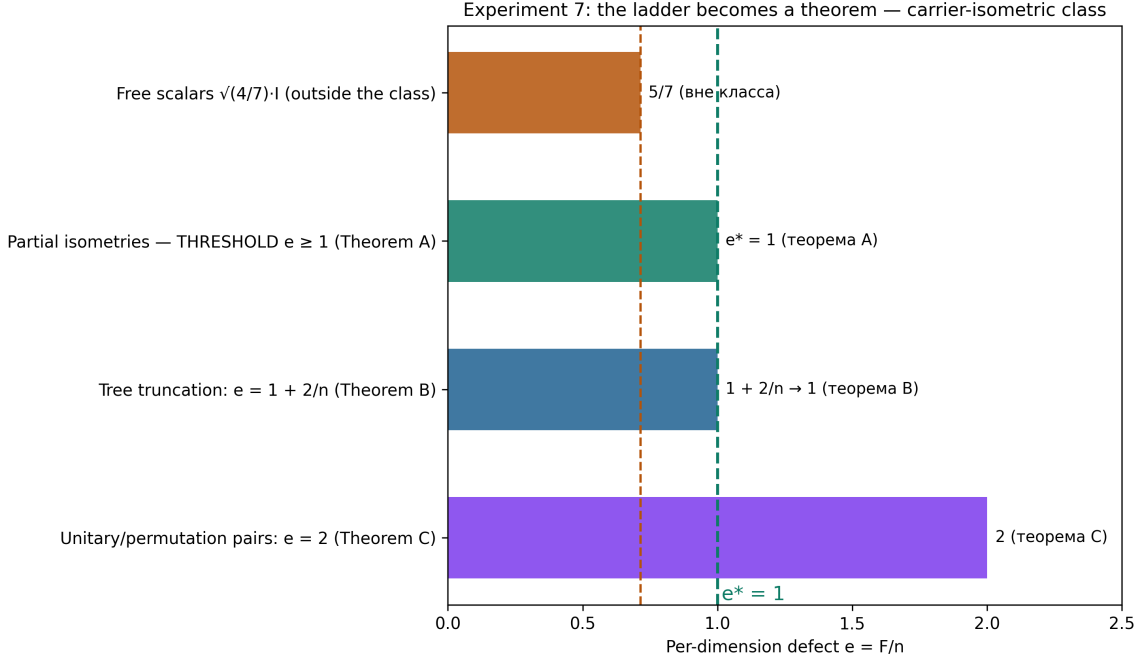


Figure 5: The ladder as a theorem: on the class of carrier-isometric pairs the floor is $e = 1$ (Theorem 5.2); the value $5/7$ is attainable only by contracting the carrier (outside the class), and unitary pairs pay exactly $e = 2$ (Theorem 5.5).

5.5 Consequence for the Sh1–Sh4 chain and the Sh.3(iii) bridge

Corollary 5.6 (class closure). *On the class of carrier-isometric pairs $\inf e = 1$; the value $5/7 < 1$ is unattainable. Hence no sequence of carrier-isometric models can drive $e \rightarrow 0$: in the Sh1–Sh4 chain the step Sh2 strengthens to $\eta \equiv 1$, and the Sh.3(iii) bridge (soficity $\Rightarrow$ matrix models with $e \rightarrow 0$) is impossible for isometric-on-carrier transports — whatever permutation data one takes, the per-dimension defect stays at or above 1.*

Remark 5.7 (the class boundary is optimal). A weighted tree $c_i S_i$ leaves the class with a quadratic penalty

$$F = (n - 1)[(c_1^2 - 1)^2 + (c_2^2 - 1)^2] + n + 2; \quad (9)$$

at $c^2 = 4/7$ this gives $e \approx 1.37\text{--}1.48$ (for $k = 3\text{--}7$) — above the class floor. Reaching $5/7$ requires contracting the *entire* carrier ($\sqrt{4/7} I$): the global minimum lives strictly outside the class, so the threshold of Theorem 5.2 is unavoidable within it.

5.6 Numerical verification

All statements were verified by the experiment `exp7_e1_threshold.py` (deterministic, fixed seeds); see Table 5. Separately, the supporting bounds $\max(0, p + q - n) \leq \operatorname{tr}(PQ) \leq \min(p, q)$ were checked on 10^4 random pairs of projections (0 violations), and the identity (7) to $5.9 \cdot 10^{-16}$.

Table 5: Numerical verification of Theorems A/B/C (Experiment 7).

Check	Scope	Max deviation
Identity $F = 3n - 2p - 2q + 3r$	24,000 random partial isometries, $n \leq 12$	$5.9 \cdot 10^{-16}$
Threshold $e \geq 1$ (Theorem A)	the same 24,000 pairs	0 violations
Equality cases $F = n$	5 configurations (n, p, q)	$8.9 \cdot 10^{-16}$
Covering formula (Theorem B)	500 random (P, Q) , $P + Q \leq I$	$7.1 \cdot 10^{-15}$
Tree: $F = n + 2$, $\text{rank}(R) = 1$	$k = 2..10$ (n up to 2047)	$2.3 \cdot 10^{-13}$
Unitary pairs $F = 2n$ (Theorem C)	200 pairs, $n \leq 32$	$2.8 \cdot 10^{-14}$
Permutation pairs $F = 2n$	200 pairs	$1.4 \cdot 10^{-14}$
Bounds on $\text{tr}(PQ)$	10,000 pairs of projections	0 violations

6 Conclusions and next steps

Assembled into one picture, the results of this session move the programme in three ways. First, the “obstruction ladder” of the Cuntz defect is no longer an empirical sequence of values but a theorem with exact arithmetic: the threshold $e \geq 1$ on the class of carrier-isometric constructions is proved (Theorem 5.2), the covering formula $F = n + 2\text{rank}(R)$ is exact (Theorem 5.4), unitary pairs pay exactly $2n$ (Theorem 5.5), and Corollary 5.6 closes the Sh.3(iii) bridge for the whole class: sofic amplification cannot come from isometric transports. Second, the DSI-4 observation acquired an analytical environment: the constant $224 = 4 \cdot 56$ is census-born (seven independent identities, cross-tower twins), the fraction $9/224$ is unique among census fractions (uniqueness lemma), and the genus family $c(g)$ with a single degeneracy at $g = 3$ (Theorem 4.1) makes the programme falsifiable: the branches $(g - 1)^{-1}$ and $(g - 1)^{-2}$ differ by a factor of 1.4 already at $g = 7$ and by nine at $g = 17$. Third, the Choptuik problem statement is fixed at both levels (physical and spectral), and its solvability is assessed honestly: numerically — solved; rigorously — major open problems (existence, uniqueness, stability of the DSS critical solution); closed form — the framework’s programme with two precise gaps (the mechanism and the 0.67% offset).

Natural next steps: (a) commit Theorems 5.2–5.5 and Corollary 5.6 to the repository as version 3 of `stability_lemma` with tests on partial isometries; (b) set up a lattice/spectral experiment on the Fricke–Macbeath surface ($g = 7$): a measurement of the DSI amplitude would discriminate branches I and III with a single number ($3/224$ versus $b_{\text{Ch}}(46)$ — a factor of 1.44); (c) a high-precision computation of λ_1 of the Klein quartic (Hejhal’s algorithm for Maass forms) and of γ by Gundlach-type perturbation theory — to test whether the monograph’s δ -expansion can close the 0.67% gap between $b_{\text{Ch}}(7)$ and γ ; (d) analytic work on the numerator $9 = 3^2$: an attempt to derive “all but one” ($N - 1$) from the cohomology of spin bundles over $\{7, 3\}$, or a proof that no such derivation exists — which would itself sharpen the honest attribution of DSI-4 as an observation.